Diphtheria

129

Introduction

- ☑ Diphtheria is a highly contagious and potentially life-threatening bacterial infection by *Corynebacterium diphtheriae* and rarely by other species, like *C. ulcerans* and *C. pseudotuberculosis*. Diphtheria is mainly affecting children <16 years of age but also can be seen in adult over age of 40 years.
- ☑ The disease transmission occurs through direct contact or inhalation of airborne oral or respiratory secretions or direct contact with skin lesions.
- ☑ In 2018, the World Health Organization (WHO) documented 16,611 reported cases out of which 8,788 were from India. In general, diphtheria is underreported in many regions, including Asia, Africa, and eastern Mediterranean countries.
- ☑ Outbreaks of respiratory diphtheria have been reported in several countries and in India between 2010 and 2016. Resurgence of cases is mainly due to incomplete vaccination.
- ☑ A survey conducted in India in 1989 identified approximately 20% of children as carriers of *C. diphtheriae* after exposure to a confirmed case of diphtheria, and 65% of the organisms were toxin producing.

Clinical Features

In Respiratory Tract Diphtheria

Clinical manifestations depend on the site of involvement.

- ☑ Most commonly site involved is tonsils or pharynx followed by nose and larynx.
- ☑ Pharyngeal and tonsillar diphtheria onset is acute and associated with low grade fever, anorexia, malaise, and pharyngitis.
- ☑ Membrane adherent to tonsils, pharyngeal wall extending to uvula and down to larynx and trachea appears in a day or two and extension of membrane depends on the immunity status of the patient. Attempts to remove the membrane are followed by bleeding.
- ☑ Cervical swelling caused by edema of underlying soft tissue and enlarged lymph nodes gives appearance of bull neck. Edema is brawny, pitting, warm on touch, nonerythematous, and tenderit can result in airway compromise. Toxins are released and further can lead to subsequent toxin mediated complications in distant organs.

Clinical Features

In Respiratory Tract Diphtheria

- ☑ Nasal diphtheria is a mild disease. It starts with common cold and is characterized by mild rhinorrhea and a lack of systemic symptoms. Later on, nasal discharge becomes serosanguinous and then mucopurulent. This form of disease is commonly seen in infant.
- ☑ Symptoms of diphtheria infection include low grade fever, anorexia malaise, pharyngitis, hoarseness, stridor, dyspnea, and a croupy cough. Toxicity and signs of toxemia generally are less iprominent. Suprasternal, subcostal, and supraclavicular retractions resulting to severe laryngeal obstruction, which can cause mortality unless intervened.

Differential Diagnosis

It includes exudative pharyngitis caused by *Streptococcus pyogenes*, acute epiglottitis, infectious mononucleosis, and adenovirus.

Cutaneous Diphtheria

Cutaneous diphtheria is typically a nonprogressive, indolent infection manifested as a superficial, ecthymic, and nonhealing ulcer with a gray-brown membrane. The extremities are more commonly affected than the trunk or head. Pain, tenderness, erythema, and exudate are all common symptoms. It is possible to experience local hyperesthesia or hypoesthesia.

Infection at Other Sites

Other mucocutaneous locations, such as ear (otitis externa), the eye (purulent and ulcerative conjunctivitis affecting mostly palpebral areas), and genital tract (purulent and ulcerative vulvovaginitis) are infrequently infected by *C. diphtheriae*.

Toxic Myocardiopathy

- ☑ Toxic myocardiopathy occurs in approximately 10–25% of patients with diphtheria and is responsible for 50–60% of deaths. As pharyngeal disease improves, the first signs of cardiomyopathy are usually noticed in the second to third week of sickness. Cardiomyopathy, on the other hand, might emerge as early as first week of sickness and as late as sixth week.
- ☑ Electrocardiogram (ECG) shows disproportionate tachycardia, prolonged PR interval, or abnormalities of ST-segment and T-wave, while an echocardiography frequently shows left ventricular dysfunction. Some may have cardiac failure or arrhythmias such as atrioventricular (AV) dissociation. Most of the myocardial complications are temporarily associated with the disease.

Toxic Neuropathy

- ☑ The onset of neurologic manifestations develop in multiphasic manners that correspond to primary illness. Hypoesthesia and local paralysis of the soft palate mostly occurs within 2–3 weeks after the onset of oropharyngeal inflammation.
- ☑ Weakness of the posterior pharyngeal, laryngeal, and facial nerves may then occur, leading to nasal quality of voice, difficulty in swallowing, and a risk of death due to aspiration.
- ☑ Oculomotor and ciliary paralysis are mainly caused by cranial neuropathies, which often appear in the fifth week which is manifested in the form of blurring of vision and difficulty in accommodation.
- ☑ There have been descriptions of both proximal muscular weakness of the extremities progressing distally and, more frequently, distal weakness progressing proximally.
- ☑ The duration between the onset of diphtheritic polyneuropathy and diphtheria disease varies from 10 days to 3 months. Most of the neurologic complication from diphtheria disease are bilateral and are motor rather than sensory.

Diagnostic Tests

- ☑ Isolation of the organism is needed for the clinical diagnosis. Specimens for culture should be obtained from the nares and throat or any mucosal or cutaneous lesion.
- ☑ Multiple samples to be taken from respiratory sites to improve the culture yield. If membrane is present then part of the membrane or material behind the membrane should be collected.
- ☑ Culture samples can be transported at 4°C in any transport media or in a sterile container. The obtained sample should be inoculated on a Löffler slant, a tellurite plate, and a blood agar plate.
- ☑ A modified Elek test can be used for detection of toxigenicity and it has fewer false-positive and false-negative results and yields results more rapidly compared to old Elek test.
- ☑ A rapid enzyme immunoassay is another test for detection of diphtheria toxin and it is a rapid test and takes 3 hours for diagnosis.
- ☑ Rapid testing for diphtheria toxin by polymerase chain reaction (PCR) specific for the A or B portion of the toxin gene is sensitive and absence of the toxin gene by PCR excludes the diagnosis of diphtheria.

Treatment

Diphtheria treatment involves:

- ☑ Using diphtheria antitoxin to stop the bacteria toxin from damaging the body
- ☑ Antibiotics used to stop toxin production and to eradicate the infection

Diphtheria Antitoxin

As diphtheria is a life-threatening infection and patient can deteriorate rapidly, to neutralize the toxin, a single dose of injectable diphtheria (equine) antitoxin (DAT) should be administered depending upon the clinical presentation, traveling history, and vaccination status before culture results are available.

The dose of antitoxin depends on the site and size of the membrane, duration of illness, and degree of toxic effects. The antitoxin dosage is empirical. Before toxin administration, skin sensitivity test should be done.

TABLE 1: Dosage of diphtheria antitoxin.

Category of disease	*Diphtheria antitoxin dose*
Pharyngeal or laryngeal disease < 48 hours	20,000–40,000 U
Nasopharyngeal	40,000–60,000 U
Severe pharyngeal or laryngeal	80,000–120,000 U

Treatment

Antimicrobial Therapy

Antibiotics should be given when diphtheria is suspected clinically and it is not a replacement for antitoxin.

TABLE 2: Antimicrobial therapy.

Antibiotics	***Doses***	***Duration***
Erythromycin (oral or IV)	40–50 mg/kg/day, maximum 2 g/day	14 days
Aqueous penicillin G (IV)	100,000–150,000 units/kg/day divided into four doses	14 days
Penicillin G procaine (IM)	25,000–50,000 units/kg/day divided into two doses	14 days

Immunization

Active immunization against diphtheria should be undertaken during convalescence from diphtheria, because disease does not necessarily confer immunity.

Cutaneous Diphtheria

Cutaneous wounds should be cleansed thoroughly with soap and water and administration of an appropriate antimicrobial agent for 10 days are recommended.

Isolation of the Hospitalized Patient

In addition to standard precautions, droplet precautions are recommended for patients and carriers with respiratory diphtheria until two cultures from both the nose and throat collected 24 hours after completing antimicrobial treatment are negative for *C. diphtheriae*.

Treatment

Control Measures

Care of Exposed People

- ☑ Whenever respiratory diphtheria is suspected or proven, local public health officials should be notified promptly.
- ☑ Close contacts of a person suspected to have diphtheria should be identified, and the following are recommended:
- ☑ For close contacts, regardless of their immunization status, the following measures should be taken: antimicrobial prophylaxis with oral erythromycin (40–50 mg/kg per day for 7–10 days, maximum 1 g/day) or a single intramuscular injection of penicillin G benzathine (600,000 U for children weighing <30 kg, and 1.2 million U for children weighing ≥30 kg and for adults).
- ☑ Follow-up cultures of pharyngeal specimens should be performed after completion of therapy for contacts proven to be carriers. If cultures are positive, an additional 10-day course of erythromycin should be administered, and follow-up cultures of pharyngeal specimens again, should be performed.
- ☑ Asymptomatic, previously immunized close contacts should receive a booster dose of an age-appropriate diphtheria toxoid-containing vaccine [DTaP (or DT), Tdap, or Td] if they have not received a booster dose of a diphtheria toxoid-containing vaccine within 5 years.
- ☑ Contacts who cannot be kept under surveillance should receive penicillin G benzathine rather than erythromycin, and if not fully immunized or if immunization status is not known, they should be immunized with DTaP, Tdap, DT, or Td vaccine, as appropriate for age.

Prevention/Vaccination

The standard schedule is three primary doses of DPT or DTaP at 6, 10, and 14 weeks and two boosters at 15–18 months and 4–5 years. The absolute contraindications to DTaP and whole-cell vaccines are same and those include history of anaphylaxis or encephalopathy following past pertussis vaccination.

TABLE 3: Vaccination for diphtheria.

Age (completed weeks/months/years)	***Vaccines***	***Comments***
6, 10, and 14 weeks	DTwP or DTaP	☑ Both can be used for primary series ☑ Immunogenicity and longevity of immune response is better with DTwP
16–18 months	DTwP B1 or DTaP B1	
4–6 years	DTwP B2 or DTaP B2	Tdap is not recommended here.
9–12 years	Tdap/Td	☑ Recommended age for Tdap is 10 years ☑ Tdap is preferred to Td followed by Td every 10 years

Further Reading

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